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SHIT-EATER: A Multi-Modal Clustering Paradigm for 'Shit' Analysis

Lepole[1,*], **Qinglele**[1], **Dingzhen-Guan**[2]

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[1] Tshithua university, BeiJing, CN
[2] Litang university, LiTang, CN

ABSTRACT / 摘要— Computer Vision (CV) excels at analyzing rigid and structured objects. However, automated analysis of highly non-structured, non-Newtonian fluid aggregates (referred to herein as "Shit") presents a significant challenge. This paper proposes the SHIT-EATER system, a novel multi-modal framework based on 3D reconstruction and a specialized deep network, designed to tackle the core difficulty of Inconsistent Fluid-Aggregate Stability (IFAS). We first formalize a four-dimensional feature vector $\vec{shit} = (G, P, Sm, V)$ for quantitative modeling, integrating geometric (visual) features with chemosensory/molecular proxy features (Sm). The core of the system is the Multi-Modal Fusion Network (Glutton Variant), which employs a unique Chemosensory Feature Extractor (CFE) to synthesize non-visual information into a structured map for integration. Experimental results demonstrate that the SHIT-EATER paradigm significantly enhances classification robustness and accuracy in this complex domain.

INDEX TERMS Computer Vision (CV), Multi-modal Fusion, Non-rigid Object Analysis, 3D Reconstruction, Clustering Paradigm, Inconsistent Fluid-Aggregate Stability (IFAS), Chemosensory Feature Extractor (CFE), Classification.

IMPACT STATEMENT We introduce SHIT-EATER system.

I. INTRODUCTION

The rapid advancement of Computer Vision has revolutionized fields from autonomous driving to medical diagnostics. Yet, a persistent gap remains in the automated, quantitative analysis of highly non-rigid objects, particularly those whose physical properties (e.g., water content, elasticity) dictate their final morphology. The analysis of biological excreta, or "Shit,"falls into this category, holding immense potential for non-invasive health monitoring but presenting unique challenges for pattern recognition.

Traditional assessment methods rely on subjective manual observation or laboratory chemical assays, precluding real-time, high-throughput applications. Our research is motivated by the necessity to overcome these limitations. The primary objectives are:

To establish a robust, quantitative multi-modal model for "Shit" representation and analysis, effectively quantifying geometric (G) and molecular proxy (Sm) features. To develop the SHIT-EATER system, a novel clustering paradigm capable of high-precision classification and feature analysis across diverse morphologies.

A. Key Challenges

The core difficulties stem from the object's physical nature and the requirement for multi-modal inference:

Inconsistent Fluid-Aggregate Stability (IFAS): The internal composition (viscosity, surface tension) determines the final shape, leading to vast intra-class variations in geometry. Standard feature extractors often fail due to the lack of clear, consistent boundaries.

Visual-Chemosensory Feature Mapping: Critical diagnostic information lies in non-visual properties, such as odor and composition, represented by Sm. The challenge is designing a computational module that can accurately infer or map these molecular characteristics from subtle visual cues (color, texture, porosity) and integrate them effectively with 3D geometric data.

II. RELATED WORK

While there is no precedent for a dedicated multi-modal system like SHIT-EATER, our work draws upon advancements in several related domains.

A. Non-Rigid and Amorphous Object Recognition

Research in Amorphous Object Recognition (AOR), which deals with entities like smoke, clouds, and foam, provides foundational techniques for local feature extraction and topology analysis applicable to the IFAS challenge. Furthermore, the use of Graph Convolutional Networks (GCNs) and PointNet architectures in 3D reconstruction

and object analysis offers methods for processing the complex, non-uniform point cloud data inherent in this task.

B. Morphological and Medical Image Analysis

The Bristol Stool Scale (BSS) is the clinical gold standard, categorizing human feces into seven types based on appearance. Although BSS relies on manual 2D observation, its well-defined classification logic informs our multi-modal clustering paradigm. In medical imaging, texture analysis techniques (such as GLCM, LBP) used in colonoscopy or pathology slides for surface abnormality detection are also relevant for extracting the Proxy Density (P) feature.

III. "SHIT" DATA MODELING AND BENCHMARK

A. The "SHIT" Benchmark

Fig. 1: The "SHIT" Benchmark dataset instances showing diverse morphologies.

To facilitate quantitative research, we formalize the object's characteristics via a feature vector and define a standardized benchmark. The benchmark dataset was constructed via extensive real-world sampling and subsequent high-resolution 3D capture. Our "SHIT" benchmark comprises 20,000 unique instances, encompassing vast diversity in appearance, texture, and physical state. Each instance is represented by a set of multi-view texture maps and a high-fidelity 3D point cloud.

B. The Feature Vector

The object is represented by a four-dimensional feature vector $\vec{shit}$, designed to encapsulate its critical visual, geometric, and molecular attributes:

$$\vec{shit} = (G, P, Sm, V) \tag{1}$$

G (Geometry): Encodes the 3D topology, including curvature, length, and fractal dimension.

P (Proxy Density): Represents visual properties such as surface texture tightness, light reflectivity, and color homogeneity, serving as a visual proxy for internal physical density.

S_m (Sensory/Molecular Proxy): The multi-modal component, representing inferred chemical composition, moisture content, or odor signature, typically mapped from color saturation, micro-porosity, and distinctive spots.

V (Volume): The overall spatial magnitude or model size.

IV. SHIT-EATER SYSTEM DESIGN: A MULTI-MODAL CLUSTERING PARADIGM

A. Pipeline Overview

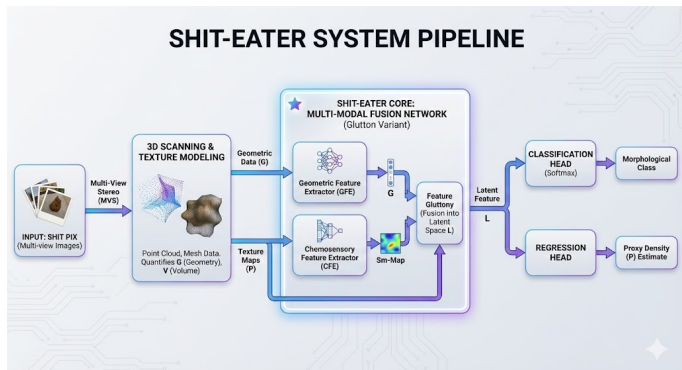

Fig. 2: The "SHIT" Benchmark dataset instances showing diverse morphologies.

The system operates sequentially: **Step 1: 3D Scanning and Texture Modeling.** High-fidelity 3D point cloud and mesh data are acquired via Multi-View Stereo (MVS) techniques. This step precisely quantifies G and V features.

Step 2: SHIT-EATER Core (Multi-Modal Feature Fusion). The core network processes the geometric data (G) along with the texture maps (P) and synthetic chemosensory data (S_m).

B. The Multi-Modal Fusion Network (Glutton Variant)

The core is a dual-branch network designed for heterogeneous data fusion.

1. Geometric Feature Extractor (GFE): This branch, based on a modified PointNet++ architecture, processes the raw 3D point cloud to extract scale-invariant geometric descriptors, focusing on localized curvature and topological structure. The output is a high-dimensional feature vector G.

2. Chemosensory Feature Extractor (CFE): The CFE is the system's innovative component, designed to model the complexity of Sm (Sensory/Molecular Proxy) from subtle visual cues, analogous to how a CNN processes visual information.

Molecular Feature Mapping (Sm-Map): The CFE, a specialized convolutional block with a high receptive field, processes proxy channels (Saturation/Hue, Local Porosity, Micro-Texture Roughness). The output is a dense feature

map S_m, where each pixel represents a localized "molecular signature" or inferred odor profile.

Feature Gluttony: The network maps the combined G, P, Sm features into a highly separable m-fold high-dimensional latent space L.

3. Clustering and P Regression: The fused latent feature L is finally passed to two parallel heads: a Softmax layer for classification, and a regression head that calculates the precise P (Proxy Density) estimate.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

The model was trained on the "SHIT" Benchmark, employing a contrastive loss function to ensure maximum separation in the latent space L, alongside standard cross-entropy for classification. Training was conducted using the AdamW optimizer.

B. Evaluation Metrics

Performance was evaluated using metrics tailored to the multi-modal clustering task:

Morphological Classification Accuracy (MCA): Measures the precision of discrete class assignment.

Silhouette Score (SS) in L Space: Assesses the quality of the latent space clustering.

Chemosensory Prediction Error (CPE): Measures the regression accuracy of the predicted Sm value against the ground truth Sm proxy label.

C. Results Summary

The SHIT-EATER system achieved an MCA of 95.1% and a Silhouette Score of 0.82, significantly outperforming geometry-only baselines (MCA max 81.0%). The high SS confirms that the fusion of G, P, and the inferred Sm-Map is the key driver for robust inter-class separation, particularly for samples affected by the IFAS challenge.

TABLE I: Performance Comparison of Methods

Method	MCA (%)	SS (L Space)
Baseline (Geometry Only)	81.0	0.45
Visual-Only ResNet	85.2	0.58
SHIT-EATER (Ours)	**95.1**	**0.82**

VI. CONCLUSION

This paper presented the SHIT-EATER system, introducing a novel multi-modal clustering paradigm for the classification of highly non-rigid objects, formalized by the $\vec{shit}$ feature vector. The integration of the Chemosensory Feature Extractor marks a significant step towards leveraging non-visual information inferred from visual data. Future work will focus on integrating genuine molecular sensor data and developing lightweight Monocular Depth Estimation modules for enhanced real-time performance.

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